Module I: Getting Started with LaTeX

Study Notes · MAT5FS112 · Syllabus units 1–4

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1. What is LaTeX?

LaTeX is a *markup language* built on top of the TeX typesetting engine. You write a plain text `.tex` file that describes the *structure* of a document (title, sections, lists, equations); the engine decides how it looks on the page.

The workflow is:

1. Write source code in a `.tex` file (any text editor, or Overleaf).
2. Compile it: `pdflatex file.tex`.
3. Open the resulting `file.pdf`.

Compilation also creates helper files (`.aux` , `.log` , `.toc`). Always read the `.log` file when an error appears; the *first* error is the one that matters.

2. Anatomy of a document

Every LaTeX file has two parts:

Preamble

Everything before `\begin{document}`. It sets the document class, loads packages and defines global settings.

Body

Everything between `\begin{document}` and `\end{document}`. This is the content you read.

2.1 The smallest document

```
\documentclass{article}
\begin{document}
Hello, world!
\end{document}
```

2.2 Document classes and options

The class decides the overall look: `article` (papers, assignments), `report` (long documents with chapters), `book`, and `beamer` (slides, Module V). Options go in square brackets:

```
\documentclass[11pt,a4paper]{article}
```

2.3 Packages

Packages add features. Load them in the preamble:

```
\usepackage[margin=2.5cm]{geometry}   % page margins
```

3. Commands, environments and special characters

A *command* starts with a backslash: `\textbf{bold}`. An *environment* is a block delimited by `\begin{name}` and `\end{name}`. Environments must be closed in the reverse order in which they were opened.

Some characters have special meaning and must be escaped:

YOU TYPE	YOU GET
`\%`	%
`\$`	$
`\&`	&
`\#`	#
`\_`	_
`\{` and `\}`	{ and }
`\textbackslash`	\
`\textasciitilde`	~

> **Comments.** A `%` starts a comment: everything after it on that line is ignored by the compiler.

4. Document structure

4.1 Title block

The title block is built from three commands in the preamble and one in the body:

```
\title{My Title}
\author{Author Name}
\date{\today}          % or \date{} for no date
\maketitle             % in the body
```

4.2 Sectioning

Sectioning commands number and title parts of the document: `\section`, `\subsection`, `\subsubsection`, `\paragraph`. Adding a star (`\section*`) removes the number and the table-of-contents entry.

4.3 Table of contents

`\tableofcontents` prints the contents page. It needs *two* compilations: the first writes the entries to a `.toc` file, the second reads them back.

4.4 Lists

Bullets with `itemize`. The `multicols` environment (from the `multicol` package) sets a list in columns:

```
\begin{multicols}{3}
\textbf{itemize} (bullets)
\begin{itemize}
  \item apples
  \item oranges
\end{itemize}
\end{multicols}
```

Numbered lists with `enumerate`; nesting is numbered differently at each level:

```
\begin{enumerate}
  \item First numbered item.
  \item Second numbered item.
    \begin{enumerate}
      \item Nested items are numbered differently.
    \end{enumerate}
\end{enumerate}
```

A description list with `description`:

```
\begin{description}
  \item[Term] an explanation of the term.
\end{description}
```

5. Bibliography

5.1 Hand-written bibliography (compiles as it is)

For a short list, `thebibliography` is enough. Each entry is introduced by `\bibitem` and cited with `\cite`. Here the citation `\cite{lamport}` and `\cite{knuth}` is resolved in this file.

```
\begin{thebibliography}{9}
  \bibitem{lamport} Leslie Lamport, \emph{\LaTeX: A Document
    Preparation System}, 2nd ed., Addison-Wesley, 1994.
  \bibitem{knuth} Donald E. Knuth, \emph{The \TeX book},
    Addison-Wesley, 1984.
\end{thebibliography}
```

Rendered:

1. Leslie Lamport, *LaTeX: A Document Preparation System*, 2nd ed., Addison-Wesley, 1994.
2. Donald E. Knuth, *The TeXbook*, Addison-Wesley, 1984.

5.2 BibTeX database (for real projects)

For larger projects, keep references in a separate `.bib` file and let BibTeX format them. The database `refs.bib` looks like this:

```
@book{lamport1994,
  author    = {Leslie Lamport},
  title     = {\LaTeX: A Document Preparation System},
  edition   = {2nd},
  publisher = {Addison-Wesley},
  year      = {1994}
}
```

The document then uses:

```
\bibliographystyle{plain}
\bibliography{refs}        % refers to refs.bib, placed at the end
```

> **Build order.** `pdflatex`, then `bibtex`, then `pdflatex`, then `pdflatex` again.

6. Index

Mark index entries in the text with `\index{term}`. Print the index at the end with `\printindex` (it needs `\makeindex` in the preamble). Here two terms are marked: LaTeX · preamble

The index is produced in three steps: `pdflatex`, `makeindex`, `pdflatex`.